

SHETH L.U.J & SIR M.V COLLEGE
Data Visualization with Power BI and Tableau

Practical No: 1

Aim:

- A) Implement the Breadth First Search algorithm to solve a given problem.
- B) Implement the Iterative Depth First Search algorithm to solve the same problem.
- C) Compare the performance and efficiency of both algorithms.

Code:

```
Artificial Intelligence practical no 1 T097.py
Artificial Intelligence practical no 1 T097.py > ...
1 from collections import deque
2
3 # 1. Representing the updated map as an adjacency list (Andheri East)
4 mumbai_map = {
5     'Chakala': ['WEH Metro', 'Andheri Kurla Road', 'JB Nagar', 'Holy Family Church'],
6
7     # Route 1 (Via WEH Metro)
8     'WEH Metro': ['Gundavali'],
9
10    # Route 2 (Via Main Road)
11    'Andheri Kurla Road': ['Gundavali'],
12
13    # Route 3 (Via JB Nagar Internal)
14    'JB Nagar': ['Tarun Bharat Society'],
15    'Tarun Bharat Society': ['Pump House'],
16
17    # Route 4 (Via Holy Family Church)
18    'Holy Family Church': ['Pump House'],
19
20    # Converging Nodes
21    'Gundavali': ['MWLU College'],
22    'Pump House': ['MWLU College'],
23
24    # Destination
25    'MWLU College': []
26 }
27
28 # A) Breadth First Search (BFS) Implementation
29 def bfs_shortest_path(graph, start, destination):
30     queue = deque([(start, [start])])
31     visited = set([start])
32     nodes_explored_count = 0
33
34     while queue:
35         current, path = queue.popleft()
36         nodes_explored_count += 1
37
38         if current == destination:
39             return path
40
41         for neighbor in graph[current]:
42             if neighbor not in visited:
43                 visited.add(neighbor)
44                 queue.append((neighbor, path + [neighbor]))
45
46     return None
47
48 # Example usage
49 start = 'Chakala'
50 destination = 'MWLU College'
51 shortest_path = bfs_shortest_path(mumbai_map, start, destination)
52
53 if shortest_path:
54     print(f"Shortest path from {start} to {destination}: {shortest_path}")
55 else:
56     print("No path found.")
57
58 # Output: Shortest path from Chakala to MWLU College: ['Chakala', 'WEH Metro', 'Gundavali', 'MWLU College']
59
60 # Time Complexity: O(V * E)
61 # Space Complexity: O(V)
```

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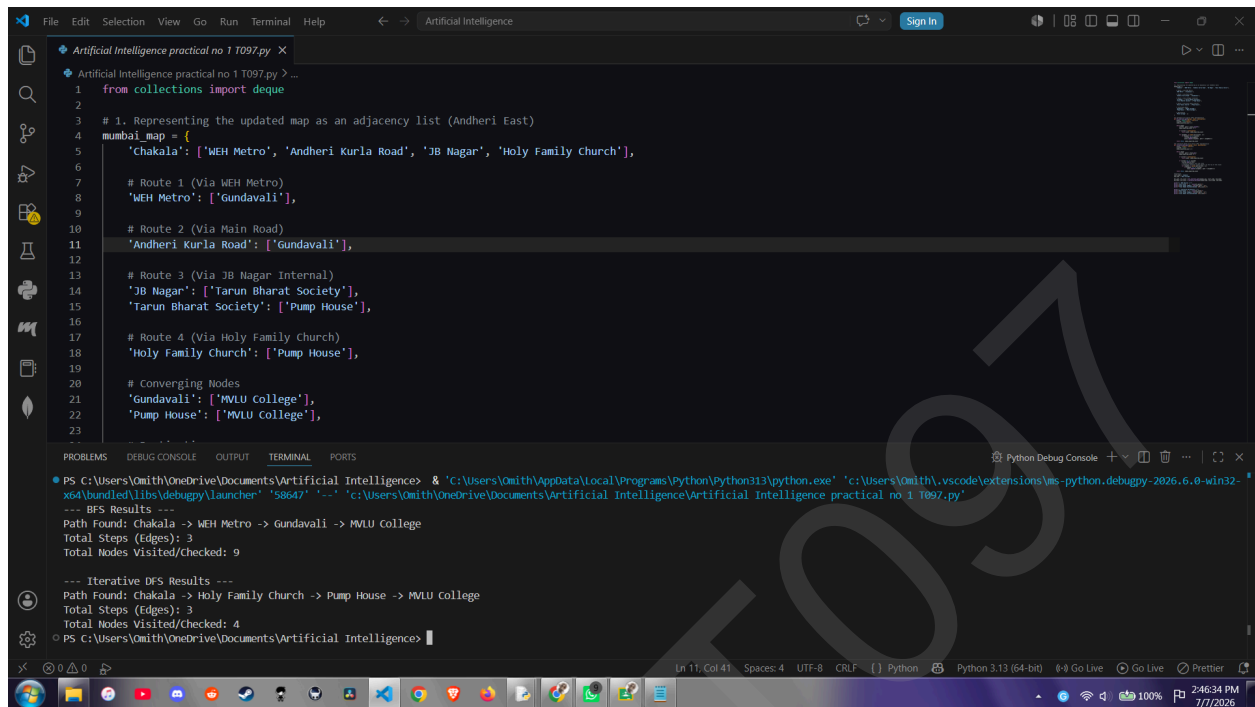
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```
Artificial Intelligence practical no 1 T097.py X
Artificial Intelligence practical no 1 T097.py > ...
29 def bfs_shortest_path(graph, start, destination):
30     if current == destination:
31         return path, nodes_explored_count
32
33     for neighbor in graph.get(current, []):
34         if neighbor not in visited:
35             visited.add(neighbor)
36             queue.append((neighbor, path + [neighbor]))
37
38     return None, nodes_explored_count
39
40 # B) Iterative Depth First Search (DFS) Implementation
41 def iterative_dfs_path(graph, start, destination):
42     stack = [(start, [start])]
43     visited = set()
44     nodes_explored_count = 0
45
46     while stack:
47         current, path = stack.pop()
48         nodes_explored_count += 1
49
50         if current == destination:
51             return path, nodes_explored_count
52
53         if current not in visited:
54             visited.add(current)
55             # Pushing neighbors onto the stack
56             # (Standard iteration pushes Route 4 to the top of the stack)
57             for neighbor in graph.get(current, []):
58                 if neighbor not in visited:
59                     stack.append((neighbor, path + [neighbor]))
60
61     return None, nodes_explored_count
62
63 # Execution
64 start_node = 'Chakala'
65 end_node = 'MVLU College'
66
67 Indexing completed.
```

```
Artificial Intelligence practical no 1 T097.py X
Artificial Intelligence practical no 1 T097.py > ...
74
75 bfs_path, bfs_count = bfs_shortest_path(mumbai_map, start_node, end_node)
76 dfs_path, dfs_count = iterative_dfs_path(mumbai_map, start_node, end_node)
77
78 print("--- BFS Results ---")
79 print(f"Path Found: {' -> '.join(bfs_path)}")
80 print(f"Total Steps (Edges): {len(bfs_path) - 1}")
81 print(f"Total Nodes Visited/Checked: {bfs_count}\n")
82
83 print("--- Iterative DFS Results ---")
84 print(f"Path Found: {' -> '.join(dfs_path)}")
85 print(f"Total Steps (Edges): {len(dfs_path) - 1}")
86 print(f"Total Nodes Visited/Checked: {dfs_count}")
87
```

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Output:



```
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17    # Route 4 (Via Holy Family Church)
18    'Holy Family Church': ['Pump House'],
19
20    # Converging Nodes
21    'Gundavali': ['MVLU College'],
22    'Pump House': ['MVLU College'],
23 }
```

PROBLEMS DEBUG CONSOLE OUTPUT TERMINAL PORTS Python Debug Console

```
PS c:\Users\Omith\OneDrive\Documents\Artificial Intelligence> & 'c:\Users\Omith\AppData\Local\Programs\Python\Python313\python.exe' 'c:\Users\Omith\.vscode\extensions\ms-python.debugpy-2026.6.0-win32-x64\bundled\libs\debugpy\launcher' '58647' '--' 'c:\Users\Omith\OneDrive\Documents\Artificial Intelligence\Artificial Intelligence practical no 1 T097.py'
--- BFS Results ---
Path Found: Chakala -> WEH Metro -> Gundavali -> MVLU College
Total Steps (Edges): 3
Total Nodes Visited/Checked: 9

--- Iterative DFS Results ---
Path Found: Chakala -> Holy Family Church -> Pump House -> MVLU college
Total Steps (Edges): 3
Total Nodes Visited/Checked: 4
PS c:\Users\Omith\OneDrive\Documents\Artificial Intelligence>
```

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